

Machine Learning approach for discovery of Collective Variable

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Abstract

A lot of high-dimensional spatiotemporal data generated from molecular dynamics simulation requires identification of few low-dimensional variables which can describe essential dynamics of a system without significant loss of informations, these low-dimensional variables often called 'collective variables' or reaction coordinates. Collective variables are used in enhance sampling method to explore the rugged free energy landscape and calculate transition probabilities between different metastable states. The choice of collective variables is not trivial for complex systems, these are many linear and nonlinear methods are used.

Introduction

Over the past few decades, the major focus of structural biology and biophysics has been to understand conformational dynamics of biological systems across a broad range of timescales with high spatial resolution (ability to visualise molecular motion). Molecular dynamics (MD) simulation acts as a computational microscope which can capture biologically relevant conformational dynamics across varied timescales with high spatial resolution. Major advances in computational hardwares and algorithms led to an ever-growing use of MD simulation in varied biological systems. A typical MD simulation generates high-dimensional spatiotemporal data which captures complex molecular motions. Simulation of increasingly larger molecules at ever increasing time scales leads to the 'curse of dimensionality' which can be summarised as follows: "as the size of the biomolecule and the simulation length increases, it also increases the number of explanatory temporal variables (e.g. H-bond distances, radius of gyration, RMSD, dihedral angles etc.) and the problem of structure discovery using temporal variables gets harder. The search of collective variable with the use of linear and nonlinear machine learning method is becoming new normal to process Molecular Dynamics simulation trajectories.

Collective variables

Molecular dynamics simulation produces high-dimensional spatiotemporal data which presents challenges for analysing. If we can find few low dimensional variables of the high dimensional data which can represent dynamics of the system without loss of much information then it will be much easier to deal with, those variables are called 'collective variable', also known as reaction coordinates. For simple system one can use guess work and prior knowledge of the system to choose CV (collective variable). For complex system it is difficult to choose a good CV, in recent years there has been attempt to find CVs using dimensionality reduction method using machine learning (ML). The most common collective variable are,

- Distance between two atoms.
- Distance between center of mass (COM) of two group of atoms.
- Dihedral angles (angle between two intersecting planes)
- RMSD (root mean square deviation)
- Radius of gyration

These are called geometric CVs.

We can also take linear or nonlinear combination of geometric CVs(e.g. dihedral angles, distances, RMSD etc..) called abstract CVs. Some methods that involves linear combination of the geometric CVs are principal component analysis (PCA), Time-lagged independent component analysis etc. and nonlinear combination are diffusion map, t-SNE etc.

Principal Component Analysis

Principal component analysis (PCA) is the most common dimensionality reduction method which transforms the high dimensional data to low dimensional data by projecting the data along the maximum variance. PCA can transform set of variables $x_1, x_2, x_3, \dots, x_N$ into low dimensional presentation $y_1, y_2, y_3, \dots, y_N$ which captures as much variance of the data as possible. In MD simulations often we want to capture the motion of molecules along largest amplitude i.e high variance. The Data set $\mathbf{y}$ can be written as a linear combination of $\mathbf{x}$

$$y_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + \dots + w_{N1}x_N$$

where y_1 is one of the principal component (PC) and $w_{11}, w_{21}, \dots, w_{N1}$ are coefficients of the vector w_1 such that $\sum_{i=1}^N w_{i1}^2 = 1$.

Steps to find PC's

- Generate mean free version of the input data of $\mathbf{x}$.
- Compute the covariance matrix $C = \frac{1}{N} \tilde{\mathbf{x}}^T \tilde{\mathbf{x}}$, where $\tilde{\mathbf{x}}$ is the mean free version of $\mathbf{x}$.
- Find the eigenvalue and eigenvector of covariance matrix C .
- The eigenvector corresponds to the largest eigenvalue captures the largest variance in the data.
- generate PC's $\mathbf{y}$ by transforming the original atomic coordinates $\mathbf{x}$ via projection matrix $\mathbf{w}$.

Notice that the number of principal component (PC) are equal to the number of dimension of our original data but we choose few PC's which corresponds to the eigenvectors of first few large eigenvalue as those eigenvectors are enough to represent the dynamics of the system without much loss. Principal components are independent to each other (as eigenvectors are orthogonal to one another because of the covariance matrix is symmetric in nature) and sorted by their variance ($PC_1 > PC_2 > PC_3, \dots$).

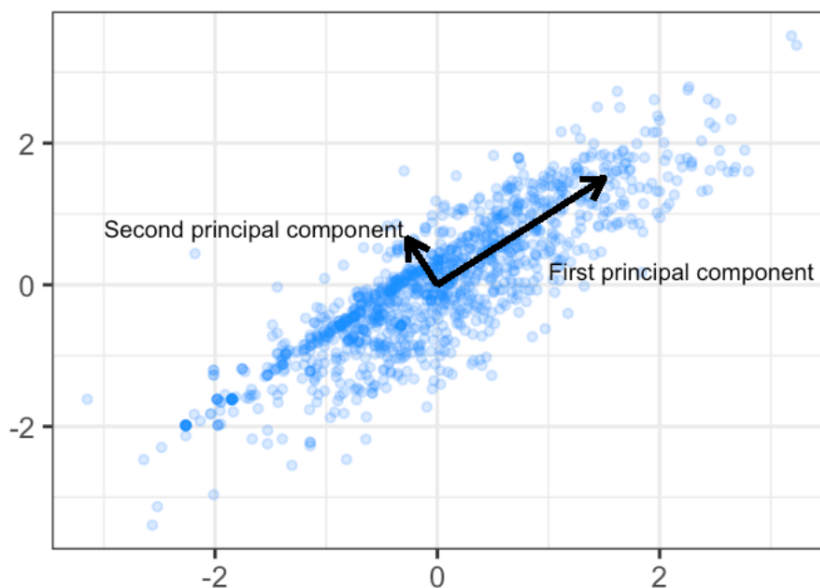


Figure 1: PCA1 and PCA2 of a high dimensional data set

It has been argued that PCA can't capture the slow conformational degrees of freedom of MD simulations. So we need to look elsewhere to get those variables that are important for slow process.

Time-Lagged Independent Component Analysis

Independent component analysis is a dimensionality reduction method which transforms a set of variables (e.g distances, RMSDs, dihedral angles etc.) into maximally independent components (IC's). The time-delayed signal can be called statistically independent if all the time-delayed correlations are zero (second-order ICA). In Molecular dynamics simulation second-order ICA is known as time-lagged independent component analysis (TICA).

Let $[X_t]_{i=1}^{N_f}$ be multidimensional time series of MD simulation data where each frame is a column vector of d-dimensions for a system. The X_t is a frame at time t and N_f is the total number of frames of MD trajectories. we want to find a projection vector $|\alpha_1\rangle$ such that the function $f(|\alpha_1\rangle)$ maximizes, which is the autocorrelation function of the projection of $|X_t\rangle$ onto $|\alpha_1\rangle$,

$$f(|\alpha_1\rangle) = \frac{\mathbb{E}[\langle\alpha_1 | X_t\rangle \langle\alpha_1 | X_{t+\Delta t}\rangle]}{\mathbb{E}[\langle\alpha_1 | X_t\rangle \langle\alpha_1 | X_t\rangle]}$$

where $\mathbb{E}$ is the expectation operator and Δt is the lag time between two time series data. Since the inner product is symmetric we can write,

$$f(|\alpha_1\rangle) = \frac{\langle\alpha_1 | C(\Delta t) | \alpha_1\rangle}{\langle\alpha_1 | \Sigma | \alpha_1\rangle}$$

Here, $C(\Delta t)$ is called time lagged correlation matrix and Σ is called the covariance matrix respectively. It can be shown that the solution of the equation also a solution of generalised solution of the eigenvalue problem

$$C(\Delta t)\alpha_1 = \lambda\Sigma\alpha_1$$

Subsequent projections, $a_2...a_d$, can be similarly found by requiring that the an solution be uncorrelated with all solutions before it. An attractive property of the above formalism is that it generalizes to multiple orthogonal reaction coordinates (TICs), ordered by decreasing time scales.

Steps to find TIC's

- Compute time-lag correlation matrix, $C(\Delta t)$, and covariance matrix Σ , from multidimensional time series data.
- Solve $C(\Delta t)\alpha = \lambda\Sigma\alpha$
- Select the top few eigenvectors (TICs) with the largest eigenvalues.

In the work of Ahalawat and co-workers [7], they have used TICA to find the optimal collective variable for GB1 β -hairpin. In that work, the two TICs with largest eigenvalues (TIC1, TIC2) were chosen for the analysis. The TIC1 and TIC2 are linear combination of the input collective variable (geometric CVs) and represented as

$$\begin{aligned} TIC1 &= C_{NBH}X_{NBH} + C_{NC}X_{NC} + C_{Rg}X_{Rg} + C_{RMSD}X_{RMSD} + C_{NW}X_{NW} \\ TIC2 &= C'_{NBH}X_{NBH} + C'_{NC}X_{NC} + C'_{Rg}X_{Rg} + C'_{RMSD}X_{RMSD} + C'_{NW}X_{NW} \end{aligned}$$

Here, TIC's are leniar combination of five geometric CVs namely NBM (Number of hydrogen bonds), NC (Native contact), Rg (Radius of gyration), RMSD (Root-mean-square deviation), NW (Number of water molecules).

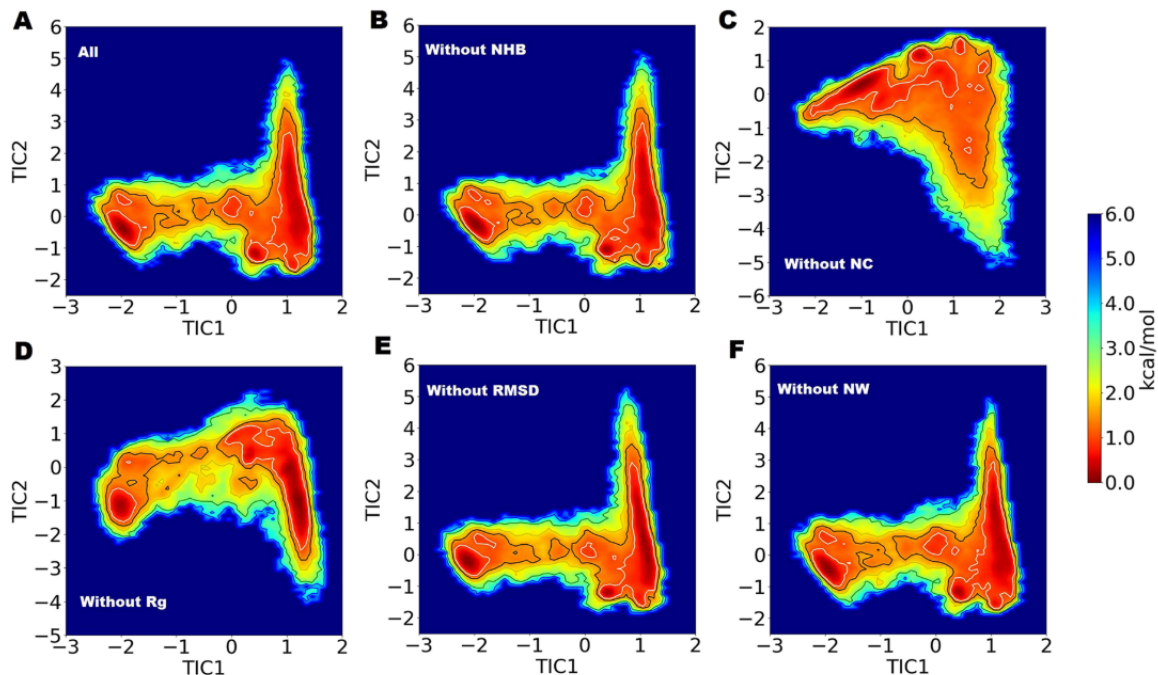


Figure 2: Two-dimensional free energy surface along the first two slowest dimensions.(a) considering all the functions: NHB, NC, Rg, RMSD, NW, after dropping (b) NHB, (c) NC, (d) Rg, (e) RMSD, (f) NW from the optimal CV. NC and Rg show the largest effect on the free energy landscape whereas others show a minimal effect.

In the free energy landscape figure[2] it can be seen that when Rg and NC are not in the linear combination of TIC1 and TIC2 the free energy deforms the most than the other cases. It can be concluded NC and Rg contribute the most toward TIC1 and TIC2 although there are significant contribution from other geometric CVs. It can be seen from figure[3] which clearly that largest contribution of Rg in TIC1 and Nc in TIC2 respectively.

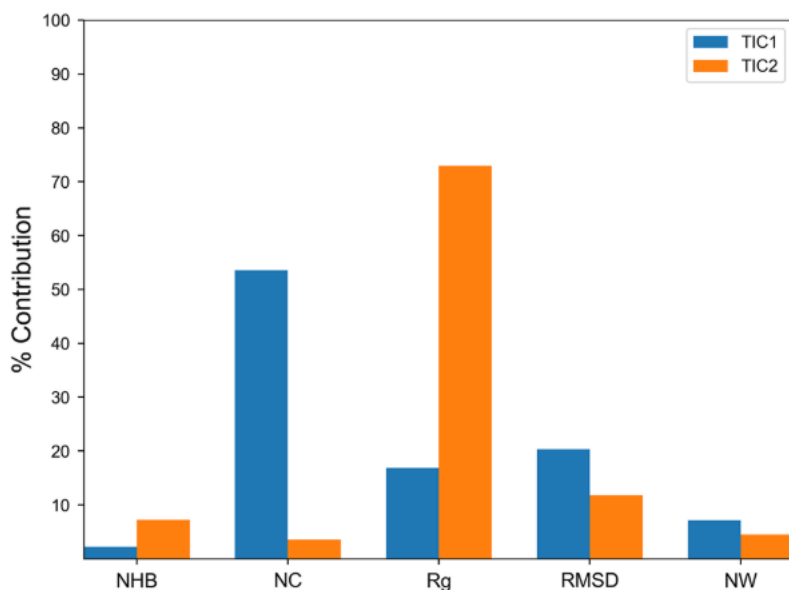


Figure 3: Percent contributions of the geometric input CVs to the first two slowest dimensions (TIC1 and TIC2).

Autoencoder

Autoencoder (AE) is a nonlinear dimensionality reduction technique which takes high dimensional data as input and learn a compressed representation (latent space) that represents the dynamics of the system, it is done by neural network architectures. Autoencoders consist of the input layer, the output layer, and one or multiple hidden layers, in between the input and output layers with a bottleneck in the middle. The layer structure is usually symmetric, with the first half including the bottleneck referred to as the encoder, while the second half is called a decoder. The input layer and output layer will have the same number of nodes, while the intermediate hidden layers in the encoder and decoder part often have fewer number of nodes, thereby forcing the network to reduce the dimensions and, consequently, to learn a lower dimensional, compressed representation in the latent space from the present dataset with a minimal regression error.

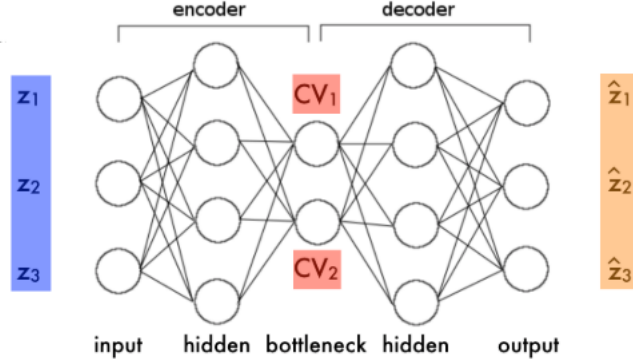


Figure 4: schematic representation of an autoencoder

In the work [5] use of autoencoder has been shown for a system of 32-bead polymer chain. Instead of any conventional collective variables as input CV all pairwise distances between atoms were given input dimensions in the autoencoder. It can be seen in figure[5] different conformation of polymer A, B, C, D. The collective variable that comes from autoencoder matches with the radius of gyration (R_g) which is a well known collective variable for this kind of system.

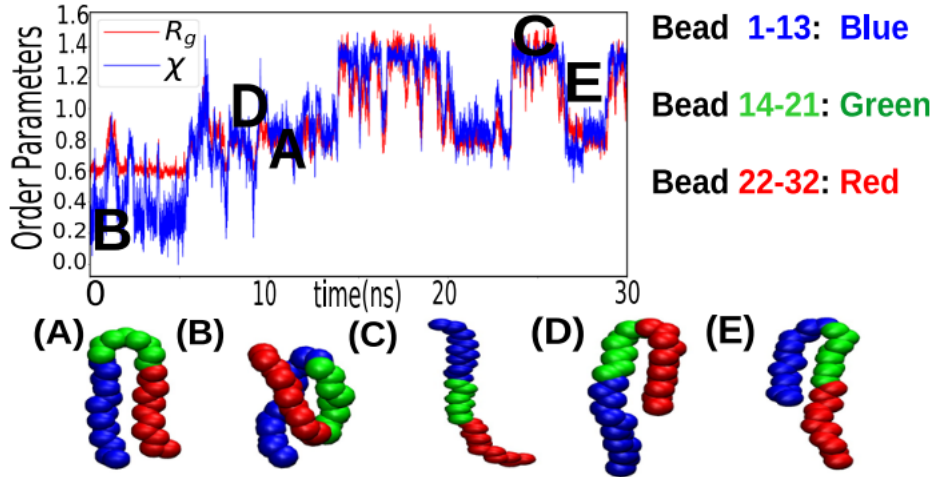


Figure 5: autoencoder captures the time-fluctuation of different conformations for a 32-bead polymer system. The time-fluctuation of R_g and ξ dimension is shown here. A, B, C, D, and E are the different conformations of the polymer.

Conclusion

In this report we have discussed about collective variables, what are those, how it is useful, how to find them. Mainly three methods are discussed PCA, TICA, AE among these first two are linear and last one is nonlinear dimensionality method. A good machine learning model for CV discovery combined with an enhanced sampling framework should sample the rare events and slow dynamics of the system. Recent metadynamics based investigation with geometric CVs (RMSD, dihedral angles etc..) highlighted the challenge of sampling slow dynamics. In order to live up to the hype, machine learning based abstract CVs should do better in sampling such rare events and predict kinetics.

References

- [1] C. R. Schwantes and V. S. Pande, J. Chem. Theory Comput. 9, 2000 (2013).
- [2] A. L. Ferguson, A. Z. Panagiotopoulos, I. G. Kevrekidis, and P. G. Debenedetti, Chem. Phys. Lett. 509, 1 (2011). 11A.
- [3] H. Sidky, W. Chen, and A. L. Ferguson, J. Phys. Chem. B 123, 7999 (2019).
- [4] G. E. Hinton, Science 313, 504 (2006).
- [5] Satyabrata Bandyopadhyay and Jagannath Mondal , "A deep autoencoder framework for discovery of metastable ensembles in biomacromolecules", J. Chem. Phys. 155, 114106 (2021).
- [6] Bhakat, S. (2021). Collective variable discovery in the age of machine learning: reality, hype and everything in between. Retrieved from <http://arxiv.org/abs/2112.03202>.
- [7] Ahalawat, N. Mondal, J. Assessment and optimization of collective variables for protein conformational landscape: GB1 -hairpin as a case study. Journal of Chemical Physics 149, (2018).
- [8] Wei Chen, Aik Rui Tan, and Andrew L. Ferguson , "Collective variable discovery and enhanced sampling using autoencoders: Innovations in network architecture and error function design", J. Chem. Phys. 149, 072312 (2018).